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Comparative evaluation of hybrid photovoltaic, wind, tidal and fuel cell clean system design for different regions with remote application considering cost

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ABSTRACT

In this paper, a comparative evaluation to optimal, cost-effective, and reliable designing of hybrid renewable and clean energy systems consisting of photovoltaic (PV), wind turbine (WT), tidal, and fuel cell (FC) energy (PV/WT/tidal/FC) with hydrogen storage (HS) is proposed. The determination of optimal system configuration is provided for three regions of Iran: Gorgan, Urmia, and Yazd with suitable weather conditions because of assessing the potential of renewable energy in these regions. The evaluation is based on real data regarding radiation, wind speed and water speed, with the objective of minimising the hybrid system net present cost (HSNPC) and satisfying the reliability constraint, i.e. the load deficit probability (LDP). A whale optimisation algorithm (WOA) with high convergence and accuracy speed is considered to determine the optimal configuration of the hybrid system with the minimum HSNPC and cost of energy (COE) that satisfies the LDP for different regions. In this study, seven different design combinations of hybrid systems are implemented. The optimal combination is determined for any region in view of the lowest cost and best reliability, as well as the contributions from renewable energy sources and storage systems. It has been also confirmed that the WOA is superior to particle swarm optimisation (PSO) for hybrid system optimisation. The results showed that optimal combination for all regions comprises a hybrid PV/WT/tidal/FC based HS. The COE values for Gorgan, Urmia, and Yazd are found to be $0.7789, $1.0864, and $0.4477, and the LDPs for these regions are 0.0090, 0.0093, and 0.0090. Moreover, the results clarified the additional contribution of tidal energy for the Gorgan and Yazd regions, the additional PV contribution to the Urmia region, and the reduced contributions related to WT sources.

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1. Introduction

The development of industry and rising living standard have led to an increase in energy demands worldwide. Renewable energies greatly reduce the problems associated with fossil fuels and nuclear energy, such as environmental pollution (Kamyab et al., 2020). Therefore, the use of these sources of energy has been emphasised in recent years under international treaties and conventions (Klemeš et al., 2020). Photovoltaic (PV) and wind turbine (WT) energy are among the best sources of renewable energy in many countries, and various policies and incentives have been implemented to reduce energy dependence on fossil fuels (Zin et al., 2016). In addition to having massive reserves of oil and gas, Iran

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is located in one of the most northerly latitudes, in a region of Earth with proper radiation. The intensity of solar radiation in Iran is greater than the global average, and more than 280 sunny days are recorded in 90% of the regions in Iran. The capacity of the large-scale PV power plants installed in Iran in 2019 was more than 67 MW, bringing the total capacity of these power plants to approximately 222 MW.

In addition to large-scale power plants, small-scale PV power plants have recently become widely used by electricity customers (Parizad et al., 2018). According to the 'Renewable Energy and Energy Efficiency Organization', the total capacity of these power plants has reached more than 20 MW (Jahannoosh et al., 2020). In total, more than 260 MW of PV power plant energy in Iran has been exploited, accounting for more than 41% of Iran's renewable electricity production (Naderipour et al., 2020). However, the total electricity generated from renewable energy sources in Iran is less than 1% of the total electricity consumption. Owing to Iran's supportive policies for the technological development of PV energy, the cost of PV power plant components has fallen sharply in recent years. Owing to the presence of many windy areas in Iran, a suitable platform has been provided for increasing the utilisation of renewable wind energy. One of the most important projects is compilation of a wind atlas for Iran. According to the atlas, the wind power is estimated to be approximately 60,000 MW, based on detailed data obtained from 60 stations in various regions in Iran ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran, and "Meteorological Organization, Iran,"). According to the forecasts, the wind energy available in Iran is calculated at 18,000 MW, indicating the potential for wind power production and cost-effective investment in the wind energy industry. Currently, the total capacity of the wind power plants installed in Iran is 72.73 MW, which is far from the targeted yearly installation of 1690 MW and 90 MW. To achieve this objective, the Iran Renewable Energy Organization is attempting to use the private sector and foreign investors to build wind power plants. The installed capacity of Iran's electricity production is 77,000 MW, and 89% of it is generated by fossil-fuel power plants ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,"). The share of renewable energies is 11%; 10% of this energy is related to tidal power, and the rest is for other renewable energies. Off-grid electricity production is a way to power residential, commercial, industrial, and rural regions, but is often impossible, owing to geographical locations and high costs of transmission.

In such cases, the use of PV and wind energy can facilitate the development of these regions (Naderipour et al., 2019). Taking advantage of low-cost options to develop new energy services in these regions is a viable and sustainable solution. Hybrid technologies combining multiple sources of renewable energy may also be applied to an energy storage system (Ramesh and Saini, 2020). A hybrid production system is a combination of one or more energy resources with storage technologies such as batteries or fuel cells (FCs) (Marassi et al., 2020). A hybrid system based on energy sources can be very versatile and can have a variety of applications. Each of the system combinations can be used based on a geographical region, access to a variety of renewable energies, and technical and economic constraints. Therefore, the radiation patterns and wind speeds of a geographical region have a significant impact on the contributions of PV and wind units and on the costs of designing and implementing such systems, such that the cost of energy (COE) per kWh generated by these systems in different geographic regions is different (Weschenfelder et al., 2020). For this reason, before implementing hybrid systems, potential studies must be conducted to determine renewable contributions and the costs of design, as well as to determine the best combination of renewable resources to achieve the lowest cost (Naderipour et al., 2020).

Numerous studies have been conducted on the potential and design of renewable energy hybrid systems in different regions with different energy sources, objective functions, reliability constraints, software environments, and optimisation tools. In (Askarzadeh, 2017), the design of a PV/diesel system for remote applications was studied in the Kerman region of Iran, with the aim of minimising the system annual cost considering the loss of power supply probability (LPSP) constraint. The design employed a harmony search algorithm. The contribution of the PV units was higher than that of the diesel units for this region. In (Maleki and Askarzadeh, 2014), the design of a hybrid PV/WT/FC system employing an artificial bee colony method was described based on the LPSP index, and with the objective of minimising the annual cost of the system. The design was conducted for different reliability constraints. The results showed that the participation of both renewable sources (and a FC) has the lowest design cost, and that the WT/FC system has the highest cost. In (Maleki and Pourfayaz, 2015), the author(s) determined the optimal system capacity for a stand-alone hybrid PV/WT/battery in the Rafsanjan region of Iran with LPSP technology, and minimised the total annual cost using evolutionary algorithms. This study indicated that PV power contributed more than wind power. Moreover, the PV/WT/battery combination was obtained as the optimal reliable combination with the lowest cost. In (Kaabeche and Ibtiouen, 2014), the optimal design of a hybrid PV/WT/battery/diesel system was presented, with the aim of minimising the present values of costs, considering the energy deficit of the system.

The best combination with the lowest cost was the PV/WT/battery/diesel system. In (Zhang et al., 2018), a PV/WT/diesel system design was evaluated using a battery and an FC, for the purpose of projecting the life span cost reduction using simulated annealing. The results showed that applying a battery is less costly than applying an FC; however, an FC is more reliable as long-term storage. In (Sanajaoba, 2019), a firefly algorithm (FA) was applied for solving the system design problem. The best combination of the system was found, with minimised cost and satisfied reliability constraints. The results showed a feasible design for the loss of load for a maximum of 3%, and the PV/WT/battery combination was determined as the lowest-cost option. In(Eteiba et al., 2018), a PV/biomass/FC system was investigated based on cost and reliability factors. The obtained contribution from PV panels was more than that from biomass. Moreover, a combination of all sources was identified as the optimal combination. In (Maleki et al., 2016), a PV/WT system design with FC storage was investigated, aiming a project life cost minimisation and LPSP satisfaction. The results indicated that the FC enhances the reliability and that PV/WT/FC is the optimal combination. In (Ramli et al., 2018), sizing of a PV/WT/diesel system with a battery was investigated with COE minimisation and LPSP observation, including a case study for a Saudi Arabia-based differential evolution.

The results indicated that PV and diesel energy have larger and smaller contributions, respectively, to supplying the load. The design of a PV/WT/battery system was studied in (Hatata et al., 2018) by using a combined genetic algorithm and particle swarm optimisation (PSO) to minimise the net present cost and improve the reliability in the Tehran region of Iran. The results illustrated that the COE for the optimal PV/WT/battery combination was $0.50. In (Fodhil et al., 2019), a potential assessment and optimisation were conducted for a PV/diesel/battery power system in urban power supply in Algeria, with the aim of minimising energy cost, and considering the loss of load probability by using a PSO. The COE was obtained as $0.38, and the highest contribution belonged to PV energy. The design of a PV/battery system in Madrid (Spain) was described in (Grande et al., 2018).

It provided a 634 kWh demand for an electric vehicle charge station, with the aim of reducing component costs using 'Hybrid

Optimisation of Multiple Energy Resources' (HOMER) software. The obtained COE value was between $0.25 and $0.4. The design of a PV/WT/diesel/battery system for use in 13 kW daily telecommunication was evaluated for Punjab (India) in (Khan et al., 2017), with the objective of reducing system costs by using HOMER software. The COE was determined as $0.137. The design of a PV/battery/FC system for residential application in the Sarawak region of Malaysia with daily consumption of 140 kWh was studied in (Das et al., 2017), with the aim of reducing system costs using HOMER software, and with the COE equal to $0.355. The design of a PV/WT/battery system in South Korea with daily consumption of 33,954 kWh was presented in (Park and Kwon, 2016), with the aim of reducing system production costs by using HOMER. The COE was $0.326. A hybrid system was designed for the Tehran region of Iran in (Fazelpour et al., 2016) with daily consumption of 17 kWh, with an aim to minimising the cost of the system via HOMER. The COE was calculated as $0.783 for a WT/FC/battery combination, and $0.793 for a PV/WT/battery combination. The design of a hybrid system for Turkey was studied in (Kalinci, 2015), with load demand of 1875 kWh, and with an aim of reducing the system production costs using HOMER software. The COE was $1.16 for a WT/FC system, and $0.836 for a PV/WT/FC system.

A production system was designed for Malaysia (Ashourian et al., 2013) with a 496 kW load demand, using the HOMER software. The obtained COE was $1.230 for a PV/FC system, $1.104 for a PV/WT/battery system, and $1.108 for a PV/WT/FC system. The optimised design of a hybrid system in Brazil with a 23.8 kW load demand was investigated in (Silva et al., 2013), with the aim of reducing energy production costs using the HOMER software. The COE was $0.657 for a PV/battery system, and $1.351 for a PV/FC system. The optimal design of renewable and clean energy systems was presented in (Karakoulidis et al., 2011) for the Kavala region in Greece, for supplying a 48 kWh load demand and reducing annual costs using the HOMER software. The COE was calculated as $1.920 for a PV/FC system, and $0.650 for a PV/diesel/battery system.

The design of an energy production system for the Istanbul region of Turkey was provided in (Türkay and Telli, 2011) with daily consumption of 627 kW, using HOMER. The COEs for the PV/WT/FC and WT/FC combinations were $3.391 and $3.847, respectively. A PV/WT/FC system was optimised in (Moghaddam et al., 2019), with the aim of minimising the net present cost using a flower pollination algorithm for the Ardabil region in Iran. The COE for this cost-effective combination was determined as $0.52. The design of a hybrid PV/WT/diesel/battery system for Iran was presented in (Borhanazad et al., 2014), with an aim of minimising the COE in production by using a PSO algorithm. The COE was $1.87 $ for the Nahavand region, $0.32 for the Rafsanjan region, and $0.35 for the Khash region; thus, the lowest cost was for the Rafsanjan region. The PV/FC/Battery system in (Al-Sharafi et al., 2017) was optimally designed to reduce the cost of power production using HOMER software, for the Jeddah region of Saudi Arabia. The COE was equal to $1.40. Due to the fact that the cost of energy (COE) for different systems and regions is different, Therefore, the amount of COE in different studies, type of hybrid systems and the optimisation methods are investigated. Disadvantages of previous studies with high COE levels are poor input data and not using precise design indices and proper design methods.

The cost of each kW hour of load supply depends on the type of hybrid system used with different sources as well as the type of storage system. On the other hand, the conditions of climate change in each region have a very important effect on the amount of this cost. Therefore, the use of a hybrid system with different resources and a reliable storage system, as well as patterns with appropriate input potential, affect the amount of this cost. In this study, the optimal design of hybrid PV/WT/tidal/FC renewable and clean energy systems with hydrogen storage (HS) is investigated for three geographical regions of Iran (Gorgan, Urmia, and Yazd) with different potentials for radiation and wind speed. The objectives are to minimise the hybrid system net present cost (HSNPC) and to satisfy the reliability constraint, i.e. the load deficit probability (LDP). The optimal designs for different combinations of hybrid systems (seven combinations) are presented based on a whale optimisation algorithm (WOA) (Hasanien, 2018), and the best combination is found for each region, in view of the lowest HSNPC and LDP. In addition, the contribution of each renewable energy source, HS, and COE are determined for each region.

The highlights of this manuscript are as follows:

- Optimal design of seven different hybrid systems based on tidal, PV, WT, and FC energy
- Cost-effective and reliable designs for three regions of Iran, based on real data
- Determination of the optimal combination for regions using the WOA
- COE values for Gorgan, Urmia, and Yazd are determined as $0.7789, $1.0864, and $0.4477, respectively
- LDP values for Gorgan, Urmia, and Yazd are found to be 0.9, 0.93, and 0.09, respectively
- Demonstration of the superiority of WOA as compared with PSO in optimal combination design

In Section 2, the regions under study are presented, and in Section 3, the hybrid clean energy system operation is described. In Section 4, the modelling of the hybrid system components is presented. In Section 5, the optimisation method is presented, along with its implementation in problem design. Simulation results are investigated in Section 6, and conclusions are presented in Section 7.

2. Regions under study

Iran has a significant ability to deploy a variety of clean energy resources. In this study, the meteorological data of three regions of Iran with different weather conditions are used to design a hybrid PV/WT/tidal/FC system. These regions include Gorgan, Urmia, and Yazd. Gorgan is located at latitude 37°24′ (longitude) and longitude 55°15′ (east longitude), and is 52 m above sea level ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,"). It has mild weather conditions and good potential for solar radiation, but less potential for wind speed. Urmia is located at latitude 31°89′ (longitude) and longitude 54°35′ (east longitude), and is 1332 m above sea level ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,"). Urmia has relatively mild weather conditions in spring, is very cold in the autumn and winter, and has relatively good radiation in the summer. Yazd is located at latitude 37°54′ (longitude) and longitude 45°07′ (east longitude), and is 1200 m above sea level ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,"). This region is relatively warm in summer and cold in winter. The solar radiation and wind speed potential in different regions in Iran are illustrated in Figs. 1 and 2, respectively. The regions of Gorgan and Urmia have relatively good radiation, but the wind speed in these areas is not ideal. The Yazd region has excellent radiation potential, and good wind speed.

In this study, the optimal design of a stand-alone PV/WT/tidal/FC system for supplying a yearly load demand is examined, with the objective of minimising the HSNPC while considering reliability constraints. The main purpose of this study is to calculate the COE value per kWh supply of load demand for system different combinations in different regions, to provide a similar load demand. Owing to the different climatic conditions of each region, the

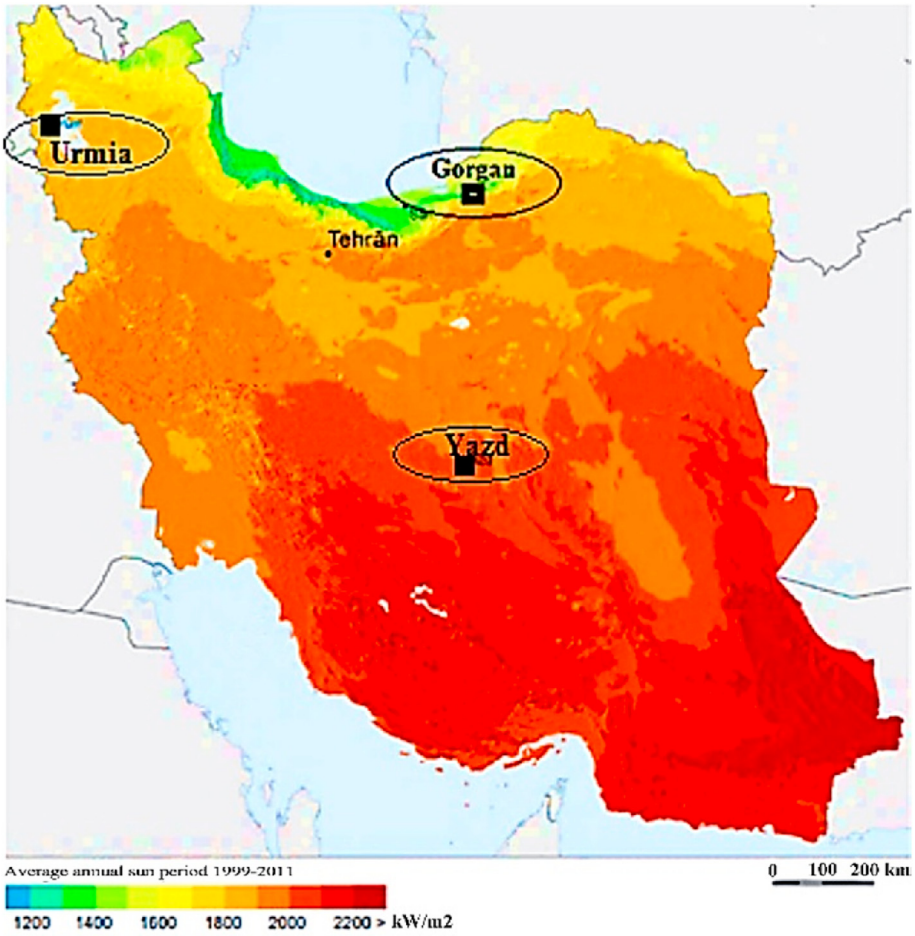


Fig. 1. Solar radiation of different regions in Iran ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,").

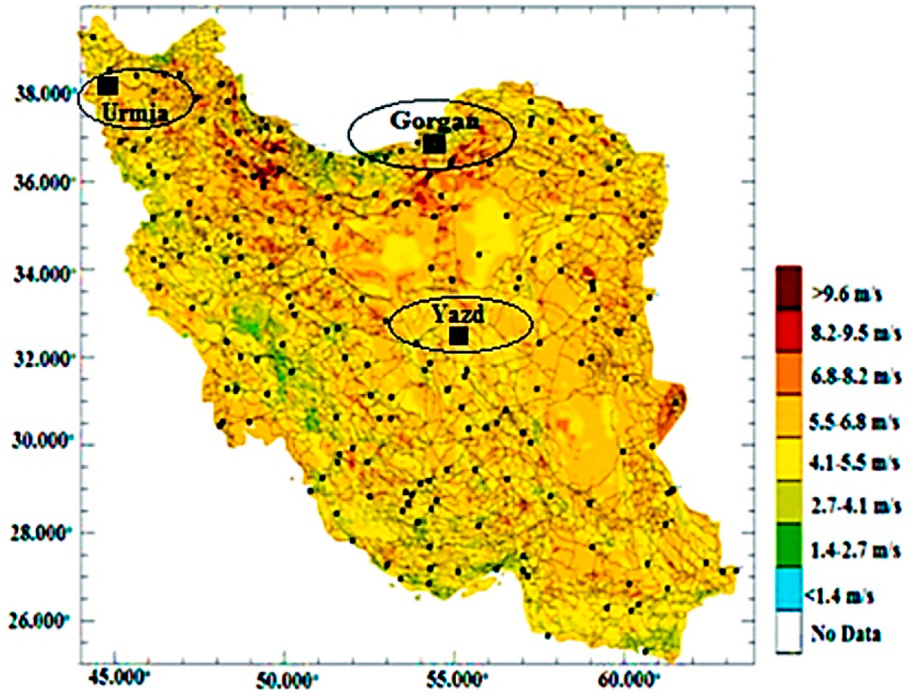


Fig. 2. Wind speed of different regions in Iran ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,").

conditions of resource sharing, and therefore COE values, appear to be different. Put another way, the patterns of radiation and wind speed affect the COE values.

3. Hybrid system energy production operation

The energy production system studied herein is a PV/WT/tidal/FC system includes tidal, PV, WT, electrolyser (EL), HS, FC, and inverter (INV) energy, as depicted in Fig. 3. The tidal, PV, and WT modules are production units, and an FC based on EL and HS is used as a storage to compensate for the power oscillations of the production units and enhance load reliability.

The hybrid PV/WT/tidal/FC system operation is presented as follows:

The total power generated by tidal/PV/WT sources is injected into the load demand.

a) $P_{HRS}(t) = (P_{Tidal}(t) + P_{PV}(t) + P_{WT}(t)) = \frac{P_{Load}(t)}{\eta_{Inv}}$

The power generated by the tidal/PV/WT sources is greater than the load demand. The extra power is injected into the EL to produce hydrogen.

b) $P_{HRS}(t) = (P_{Tidal}(t) + P_{PV}(t) + P_{WT}(t)) \geq \frac{P_{Load}(t)}{\eta_{Inv}}$

The power generated by the tidal/PV/WT sources is greater than the load demand. The extra power is injected into the EL to produce hydrogen.

c) $P_{HRS}(t) = (P_{Tidal}(t) + P_{PV}(t) + P_{WT}(t)) \leq \frac{P_{Load}(t)}{\eta_{Inv}}$

The power generated by the tidal/PV/WT sources is less than the load demand (Zhang et al., 2019). The load deficit is supplied by the FC. In this state, the LDP is defined as the reliability index, owing to the insufficient supply of load demand.

4. Hybrid photovoltaic (PV)/wind turbine (WT)/tidal/fuel cell (FC) system modelling

The hybrid PV/WT/tidal/FC system consists of tidal, PV, WT, EL, HS, FC, and INV sources, and models of each component are formulated as follows.

4.1. Tidal turbine model

The performance of tidal systems is similar to that of hydropower plants, which has led to the growth and development of their use. The tidal current flow is used for generating power. In this study, horizontal axis turbines are used. The tidal-produced power, considering cut in, cut out, and rated water flow is calculated as follows (Das et al., 2016):

$$P_{Tidal} = \begin{cases} 0 & ;V_T \leq V_{Tcutin}, V_T \geq V_{Tcutout} \\ 0.5\delta A C_P V_T^3 & ;V_{Tcutin} < V_T < V_{Tr} \\ P_{Tr} & ;V_{Tr} \leq V_T \leq V_{Tcutout} \end{cases} \quad (1)$$

Where, P_{Tidal} and P_{Tr} are the tidal output power and rated tidal power (kW), respectively, V_T is the water flow speed, $V_{Tcautin}$ is the cut-in speed, V_{Tr} is the rated speed, and $V_{Tcautout}$ is the cut-out speed (m/s). δ refers to the fluid density (kg/m^2), A is the turbine cross-sectional area (m^2), and C_P is a power coefficient.

4.2. PV production model

The PV-produced power is defined based on the solar radiation emitted to its surface, as follows (Das et al., 2017; Sanajaoba, 2019):

$$P_{PV} = \frac{SR}{SR_r} \times P_{PVr} \times EF_{MPPT} \quad (2)$$

Here, P_{PV} is the PV power, SR is the solar radiation, SR_r is the standard solar radiation (1000 W/m^2), P_{PVr} is the rated PV power, and EF_{MPPT} refers to the PV maximum point tracking efficiency.

4.3. WT production model

The WT-produced power considering the cut in, cut out, and rated wind speed is obtained as follows (Das et al., 2017; Moghaddam et al., 2019):

$$P_{WT} = \begin{cases} 0 & ;V_S \leq V_{Scutin}, V_S \geq V_{Scutout} \\ P_{PVr} \times \left(\frac{V_S - V_{Scutin}}{V_{Sr} - V_{Scutin}}\right)^m & ;V_{Scutin} < V_S < V_{Sr} \\ P_{PVr} + \frac{P_{cutout} - P_{PVr}}{P_{cutout} - P_{PSr}} \times \left(V_S - V_{Sr}\right) & ;V_{Sr} \leq V_S \leq V_{Scutout} \end{cases} \quad (3)$$

Where, P_{WT} is the wind power, V_S is the wind speed, V_{Scutin} is the cut-in speed, V_{Sr} is the rated speed, $V_{Scutout}$ is the cut-out speed, MPWG is the maximum power of the wind generator unit (kW), and P_{cutout} is the output power at the cut-out speed. The wind data is recorded at a height of 40 m, and the turbine installation height is 15 m. The wind speed at this altitude is calculated as follows (Maleki and Askarzadeh, 2014; Moghaddam et al., 2019):

$$V_{SH} = V_{Sr}\left(\frac{H_{Tower}}{H_{Tower,ref}}\right)^Q \quad (4)$$

Here, V_{SH} is the wind speed at a height of H_{Tower}, V_{Sr} is the wind speed at the height of $H_{Tower,ref}$, and the value of parameter Q is between 0.14 and 0.25 (Moghaddam et al., 2019). The parameter Q is considered 0.14 in this study.

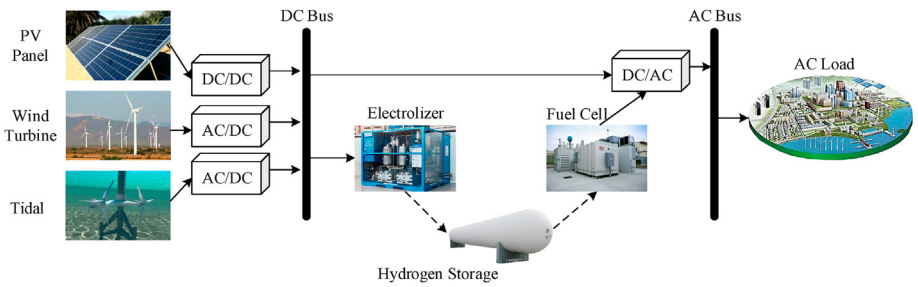


Fig. 3. Schematic of the hybrid photovoltaic (PV)/wind turbine (WT)/tidal/fuel cell (FC) system.

4.4. Electrolyser (EL) model

The EL produces oxygen and hydrogen using water electrolysis. In a condition of a hybrid system power deficit, the stored hydrogen in the HS tank is transferred to the FC to compensate for the load demand. The EL model is defined as follows:

$$P_{ELtoHS} = P_{HStoEL} \times EF_{EL} \tag{5}$$

Where, EF_{EL} is the *EL* efficiency, and $P_{EL\ to\ HS}$ is the injected power to the *EL* from the tidal, PV, and WT units.

4.5. Hydrogen storage (HS) model

The energy of the HS tank is obtained as follows (Moghaddam et al., 2019):

$$E_{HS}(t) = E_{HS}(t-1) + P_{ELtoHS}(t) \times \Delta t - P_{HStoFC}(t) \times \Delta t \times EF_{HS} \tag{6}$$

Here, $E_{HS}(t\text{-}1)$ refers to HS energy in *t-1*, P_{ELtoHS} is EL output power to HS, P_{HStoFC} is HS power injected to FC. Also $E_{HS}(t)= HHV_{H2}\times m_{Hs}(t)$ that $m_{Hs}(t)$ refers to hydrogen mass (kg) at time t, HHV_{H2} is the hydrogen heating value (Fodhil et al., 2019; Sanajaoba, 2019), and EF_{HS} is the tank efficiency.

4.6. FC model

The output FC power is calculated based on the received hydrogen, so its model is formulated as follows (Moghaddam et al., 2019):

$$P_{FCtoINV} = P_{HStoFC} \times EF_{FC} \tag{7}$$

follows:

$$P_{INVtoLoad} = (P_{FCtoINV} + P_{HStoINV}) \times EF_{INV} \tag{8}$$

Here, EF_{INV} is the INV efficiency, and $P_{HStoINV}$ is the tidal/PV/WT power transferred to the INV.

5. Methodology

In this study, a hybrid PV/WT/tidal/FC system design is investigated for three different regions of Iran, with the aim of supplying a similar load. The main goal of the system design is to minimise the HSNPC while satisfying the reliability constraint (LDP) with an optimal determination of decision variables, and optimal power from the tidal, PV, WT, EL, FC, INV, and H_2 sources. The design problem is implemented based on real data (radiation, water flow, and wind speed) of the Gorgan, Urmia, and Yazd regions.

5.1. Objective function

The optimal designing of hybrid PV/WT/tidal/FC system is presented for three regions of Iran (Gorgan, Urmia, and Yazd) with the objective of the hybrid system net present cost (HSNPC) minimisation. The HSNPC includes capital cost, maintenance cost and replacement cost of hybrid system components. The HSNPC is defined as follows (Moghaddam et al., 2019; Askarzadeh, 2017):

$$MinHSNPC = HSNPC_{cap} + HSNPC_{main} + HSNPC_{rep} \tag{9}$$

The capital cost of the hybrid system is presented as follows:
The maintenance cost of the hybrid system is defined by

$$HSNPC_{cap} = HSNPC_{Tidalc} \times N_{Tidal} + HSNPC_{PVc} \times N_{PV} + HSNPC_{WTc} \times N_{WT} + HSNPC_{ELc} \times N_{EL} + HSNPC_{HSc} \times N_{HS} + HSNPC_{FCc} \times N_{FC} + HSNPC_{INVc} \times N_{INV} \tag{10}$$

$$HSNPC_{main} = \varphi \times (HSNPC_{Tidalm} \times N_{Tidal} + HSNPC_{PVm} \times N_{PV} + HSNPC_{WTm} \times N_{WT} + HSNPC_{ELm} \times N_{EL} + HSNPC_{HSm} \times N_{HS} + HSNPC_{FCm} \times N_{FC} + HSNPC_{INVm} \times N_{INV} \tag{11}$$

In the above, EF_{FC} refers to the FC efficiency.

The replacement cost of the hybrid system is presented as follows:

4.7. Inverter (INV) model

The injected power to the load by the INV is formulated as

$$HSNPC_{rep} = \sum_{n=1}^{y} \frac{1}{\left(1+\left(\frac{ni-ai}{1+ai}\right)\right)^{n.L}} \times \left(HSNPC_{Tidalr} \times N_{Tidal} + HSNPC_{PVr} \times N_{PV} + HSNPC_{WTr} \times N_{WT} + HSNPC_{ELr} \times N_{EL} + HSNPC_{HSr} \times N_{HS} + HSNPC_{FCr} \times N_{FC} + HSNPC_{INVr} \times N_{INV} \right) \tag{12}$$

Annual payments present value factor is defined by

$$\varphi = \frac{\left(1+\left(\frac{ni-ai}{1+ai}\right)\right)^{ul}-1}{\frac{ni-ai}{1+ai}\left)1+\left(\frac{ni-ai}{1+ai}\right)\right)^{ul}} \tag{13}$$

In the above, $HSNPC_{cap}$, $HSNPC_{main}$, and $HSNPC_{rep}$ are the capital, maintenance, and replacement costs, respectively. N_{Tidal}, N_{PV}, N_{WT}, N_{EL}, N_{Tank}, N_{FC} and N_{INV} are the numbers of tidal, PV, WT, EL, tank, FC, and INV sources. $NSNPC_{Tidal}$, $NSNPC_{PV}$, $NSNPC_{WT}$, $NSNPC_{EL}$, $NSNPC_{Tank}$, $NSNPC_{FC}$ and $NSNPC_{INV}$ are the respective costs of the component units, including the maintenance cost (with index m) and the replacement cost (with index r). ni is the nominal interest, and ai is the annual inflation. ϕ refers to the annual payments present value factor, and $\sum_{n=1}^{ncr}\frac{1}{\left(1+\left(\frac{ni-ai}{1+ai}\right)\right)^{ul.ncr}}$ is the net present value of constant payments. In addition, ul and ncr represent the project lifespan and the number of replacements of components, respectively.

5.2. Constraint

In this study, the LDP is selected as a reliability constraint for the technical evaluation of the hybrid system. The LDP ranges between 0 and 1. A value of 0 means that the entire load is supplied, and a value of 1 means that the entire total load is not supplied. The LDP is presented in terms of the deficit power of the hybrid systems and the inadequate HS for supplying the load fully, as follows (Moghaddam et al., 2019; Askarzadeh, 2017):

$$LDP = \frac{\sum_{t=1}^{T}\left[P_{Load}(t) - P_{HStoINV(t)-E_{\mathrm{Tan}\ k}(t)}\right]}{\sum_{t=1}^{T}\left[P_{Load}(t)\right]} \tag{14}$$

The reliability index is defined as follows:

$$LDP \le LDP^{\max} \tag{15}$$

Here, LDP^{max} is the maximum value of the LDP. The constraints related to the maximum and minimum sizes of decision variables are presented as follows:

$$P^{\min}_{Tidal,PV,WT,EL,FC,INV} \le P_{Tidal,PV,WT,EL,FC,IN} \le P^{\max}_{Tidal,PV,WT,EL,FC,INV} \tag{16}$$

$$M^{\min}_{HS} \le M_{HS} \le M^{\max}_{HS} \tag{17}$$

$$E^{\min}_{HS} \le E_{HS} \le E^{\max}_{HS} \tag{18}$$

In the above, $P^{\min}_{Tidal,PV,WT,EL,FC,INV}$ and $P^{\max}_{Tidal,PV,WT,EL,FC,INV}$ are the minimum and maximum power for the combination of the tidal, PV, WT, EL, FC, and INV sources, respectively. $M^{\min}_{HS}$ and $M^{\max}_{HS}$ refer to the minimum and maximum hydrogen mass of the tank (kg), respectively, and $E^{\min}_{HS}$ and $E^{\max}_{HS}$ refer to the minimum and maximum tank energy (kWh), respectively.

5.3. An overview of the whale optimisation algorithm (WOA)

The design of the hybrid PV/WT/tidal/FC system for the regions of Gorgan, Urmia, and Yazd is conducted based on the WOA. The structure of WOA is presented herein. The WOA is inspired based on a behaviour of foraging humpback whales named 'bubble-net feeding', as shown in Fig. 4. The whale has twice the number of spindle cells compared to an adult human, which is the main reason for their intelligence. It has been proven that whale can think, learn, judge, communicate and even have emotional relationships. But it is clear that these wall capabilities are several levels lower than human capabilities. The whale (usually the killer whale) are also able to develop their own dialect. Another interesting point is the whale's social behavior, which is mostly seen in groups. For example, the killer Whale lives most of his life in the family. The most interesting thing about the humpback whale is that it hunts in a certain way. This is the feeding behavior called the pure feeding bubble method. The humpback whale prefers to hunt in the form of a krill school or small fish near the surface. In the path of the bubbles, two manoeuvres are observed: upward spiral and in the form of double rings. In the first manoeuvre, the humpback whale dives to a depth of 12 m and then begins to form a bubble in a spiral shape, hunting around and swimming towards the surface. The next manoeuvre involves three different steps: the coral rings and the recording ring. It should be noted that here the nutrition of the pure bubble is unique. Behavior that can only be seen in the whale.

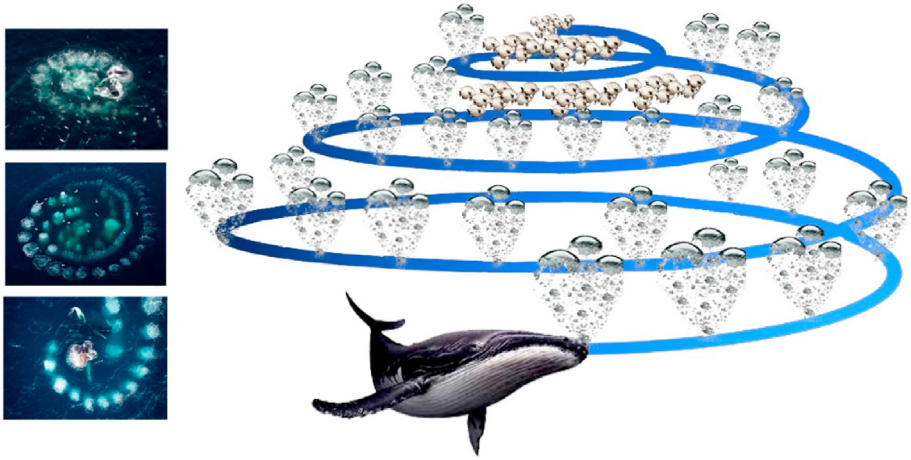


Fig. 4. Hunting behaviour of humpback whales (Mirjalili and Lewis, 2016).

5.3.1. Hunting siege

A humpback whale encircles a hunting area by identifying its location so that the best solution is considered as the nearest location to the hunting. Then, the search is performed, and the position of the humpback whale is updated. This behaviour is modelled as follows (Hasanien, 2018; Mirjalili and Lewis, 2016):

$$\vec{D} = \left| \vec{E}.\vec{X}^{*}(t) - \vec{X}(t) \right| \tag{19}$$

$$\vec{X}(t+1) = \vec{X}^{*}(t) - \vec{A}.\vec{D} \tag{20}$$

Here, t represents the current iteration, A and C are vector coefficients, X^* is the vector of the position of the best solution, and X is the hunting location vector. The vectors A and C are defined as follows (Hasanien, 2018; Mirjalili and Lewis, 2016):

$$\vec{A} = 2\vec{a}.\vec{r} - \vec{a} \tag{21}$$

$$\vec{C} = 2.\vec{r} \tag{22}$$

The parameter $\vec{a}$ is a linear vector that declines from 0 to 2 for iterations, and $\vec{r}$ is a random vector in the range of [0,1].

5.3.2. Phase of exploitation, the pure bubble attack method

The position of the humpback whale is expressed by the search agent anywhere between the original position and the current best position. The distance between the whale in position (X,Y) and the position of hunting in position (X^*,Y^*) is expressed as a spiral relation. To model this simultaneous, there is a probability of 50% to select between either the shrinking encircling structure or the spiral relation to updating the whale's position in the optimisation process. The behaviour is modelled as follows (Hasanien, 2018; Mirjalili and Lewis, 2016):

$$X(t+1) = \begin{cases} \vec{X}^{*}(t) - \vec{A}.\vec{D} & if \quad p < 0.5 \\ D'.e^{b1}.\cos(2l) + X^{*}(t) & if \quad p > 0.5 \end{cases} \tag{23}$$

Where, p is a random number in [0,1].

5.3.3. Phase of exploration, search for hunting

Based on the variety of vectors, this approach can be used to search for prey (exploration). In fact, the whale is randomly searching by position. Therefore, the search agent is considered to move away from a reference whale using random values of 1 or less than −1 value. The humpback whale search is random but is based on the position. During the exploration phase, the search position is updated randomly. This phase is modelled as follows (Hasanien, 2018; Mirjalili and Lewis, 2016):

$$\vec{D} = \left| \vec{C}.\vec{X}_{rand} - \vec{X} \right| \tag{24}$$

$$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A}.\vec{D} \tag{25}$$

In the above, $\vec{X}_{rand}$ refers to a random position vector.

In each iteration, search agents randomly select and update their position according to each search agents as the best solution. Providing an exploration and exploitation mechanism reduces a

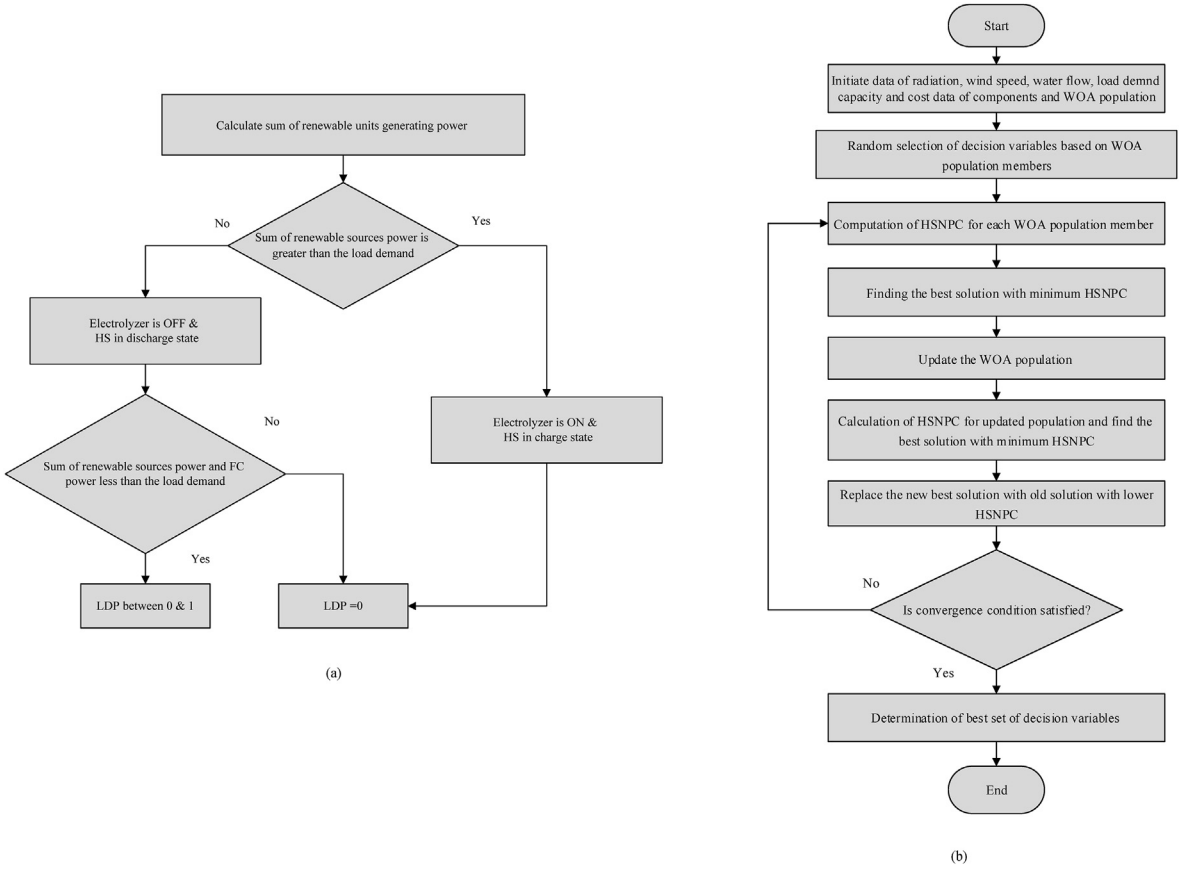


Fig. 5. Block diagram of a) hybrid system energy production operation b) whale optimisation algorithm (WOA) implementation strategy for hybrid system designing.

parameter from 2 to 0, respectively. In a random search of the agent $|A| > 1$ is selected, while the best solution is $|A| < 1$ selected to update the position of search agents. In each iteration of the WOA algorithm, search agents update their position to the best solution. Given a value of p, the algorithm switches between the spiral paths and circular motions, before finally terminating on a convergence criterion.

5.4. WOA implementation in hybrid system design

The flowchart of the WOA for the design problem is shown in Fig. 5. Its optimisation steps are as follows.

Step 1. Initiate system data, including irradiance and wind speed data of different regions, load demands, and technical and economical parameters of system components.

Step 2. Initiate the decision variables for each population of the WOA.

Step 3. Compute the HSNPC for each population of the WOA.

Step 4. Determine the best population member of the WOA based on the best HSNPC value.

Step 5. Update the population of the WOA based on (24–25).

Step 6. Evaluate the HSNPC for the updated population from Step 5.

Step 7. Determine the best solution with the lowest HSNPC at Steps 4–6, and replace the solution with a previous one.

Step 8. Evaluate the convergence conditions. If satisfied, go to step 9; if not, go to step 5.

Step 9. Select the set of decision variables with the lowest HSNPC value, and save the results.

6. Simulation results and discussion

6.1. System data

The WOA is used to design the hybrid PV/WT/tidal/FC system for the three different regions of Iran, with the objective of minimising the HSNPC while considering the LDP as the reliability constraint. The total load demand is 277.78 MW. Figs. 6–8 illustrate real data for the regions of Gorgan, Urmia, and Yazd, including data regarding the annual irradiance, wind speed, water flow, and load

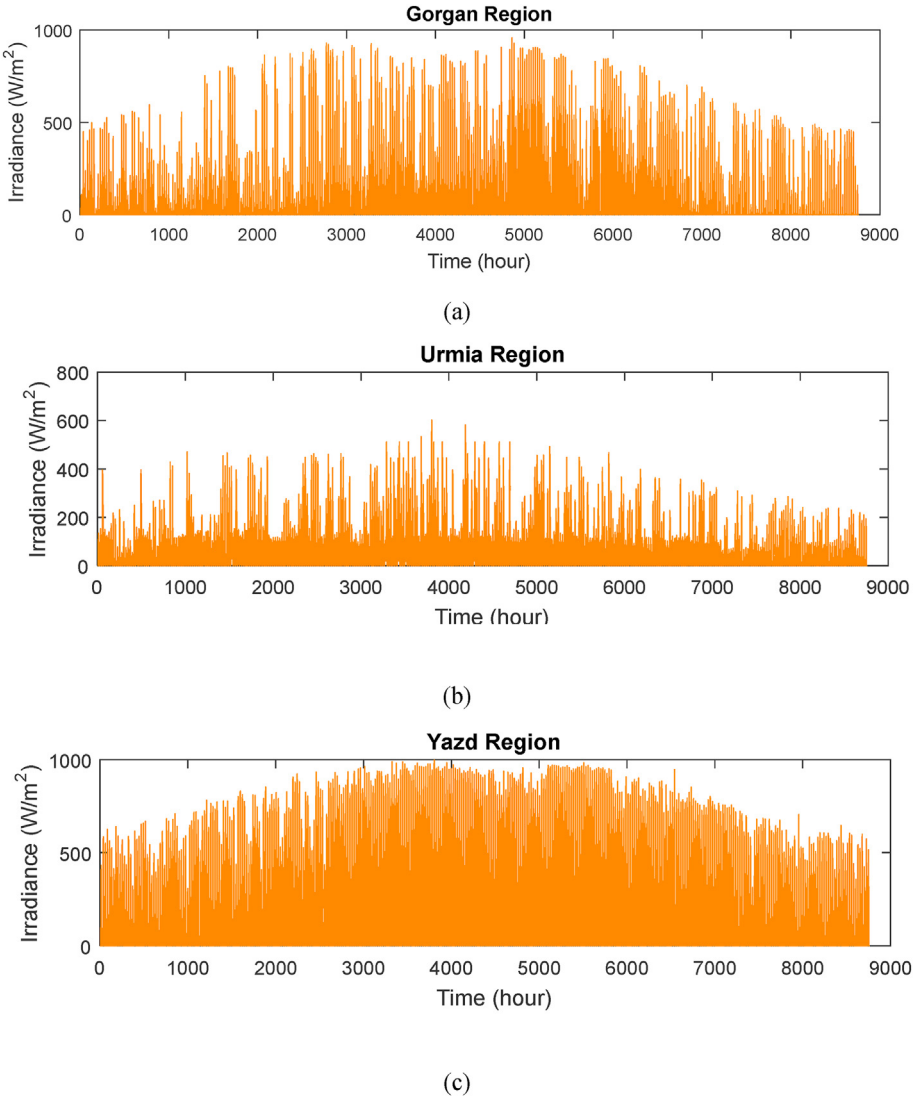


Fig. 6. Annual irradiance for different regions a) Gorgan b) Urmia c) Yazd ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,").

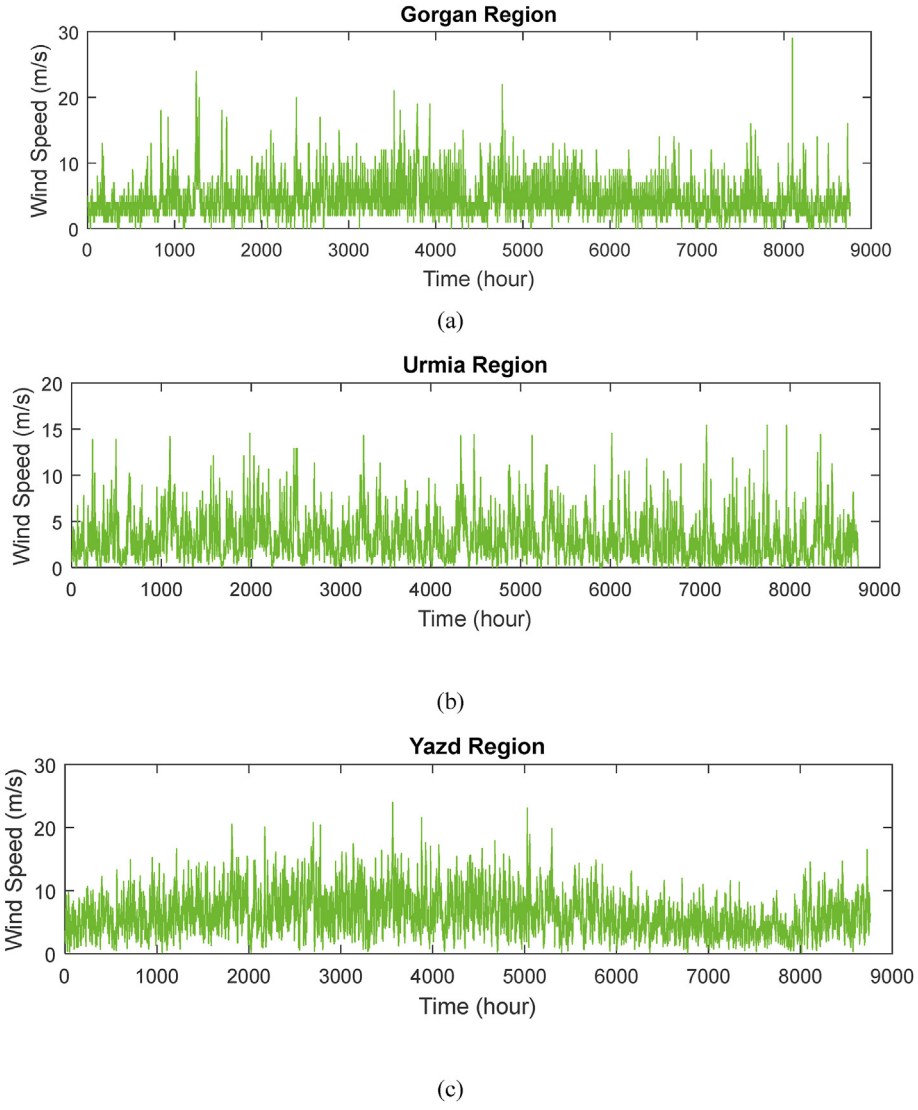


Fig. 7. Annual wind speed for different regions a) Gorgan b) Urmia c) Yazd ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,").

demand ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,"). The data regarding the capacity and the costs of system components are given in Table 1.

The rated interest is set as 9%, and the inflation rate is selected as 3% (Das et al., 2017). The WOA performance is evaluated by using PSO in the design of different combinations of hybrid systems for different regions. The number of populations and maximum iterations for the WOA is selected as 70 and 300, respectively. In this paper, the maximum iteration of the algorithm is considered 300. In each iteration of the algorithm in the maximum and minimum range of decision variables, the set of variables is determined randomly and the value of the HSNPC objective function value is determined according to selected variables by considering the LDP reliability constraint. However, according to the number of whale populations (70) and the process of updating the algorithm, the best whale with the best value of the objective function is determined. In the first iteration, the value of the objective function is obtained. Therefore, in order to achieve the best set of variables corresponding to the best objective function, these iterations are based on the extracted results and also the experience of the authors for 300 iterations to obtain the optimal set of variables and the best value of the objective function. The decision variables HSNPC, LDP, and COE are determined optimally by using WOA for the three regions of Gorgan, Urmia, and Yazd. The minimum and maximum values of the optimisation variables are 0 and 2000, respectively.

6.2. Results of optimal design for different regions

In this section, the optimal design results are presented, including the optimal contribution of components, the reliability index, and the cost of the different hybrid system combinations for different regions, with the aim of minimising the HSNPC and satisfying the $LDP^{max} = 5\%$ reliability constraint using the WOA method. The total load demand is 277.78 MW. The respective optimal power shares of tidal, PV, WT, EL, FC, INV, and tank hydrogen mass-energy are obtained for different combinations and regions. The HSNPC and LDP are also determined for the regions of Gorgan, Urmia, and Yazd.

6.2.1. Gorgan region

The results of the different hybrid system combinations for the Gorgan region are provided in Tables 2 and 3. Comparison of the

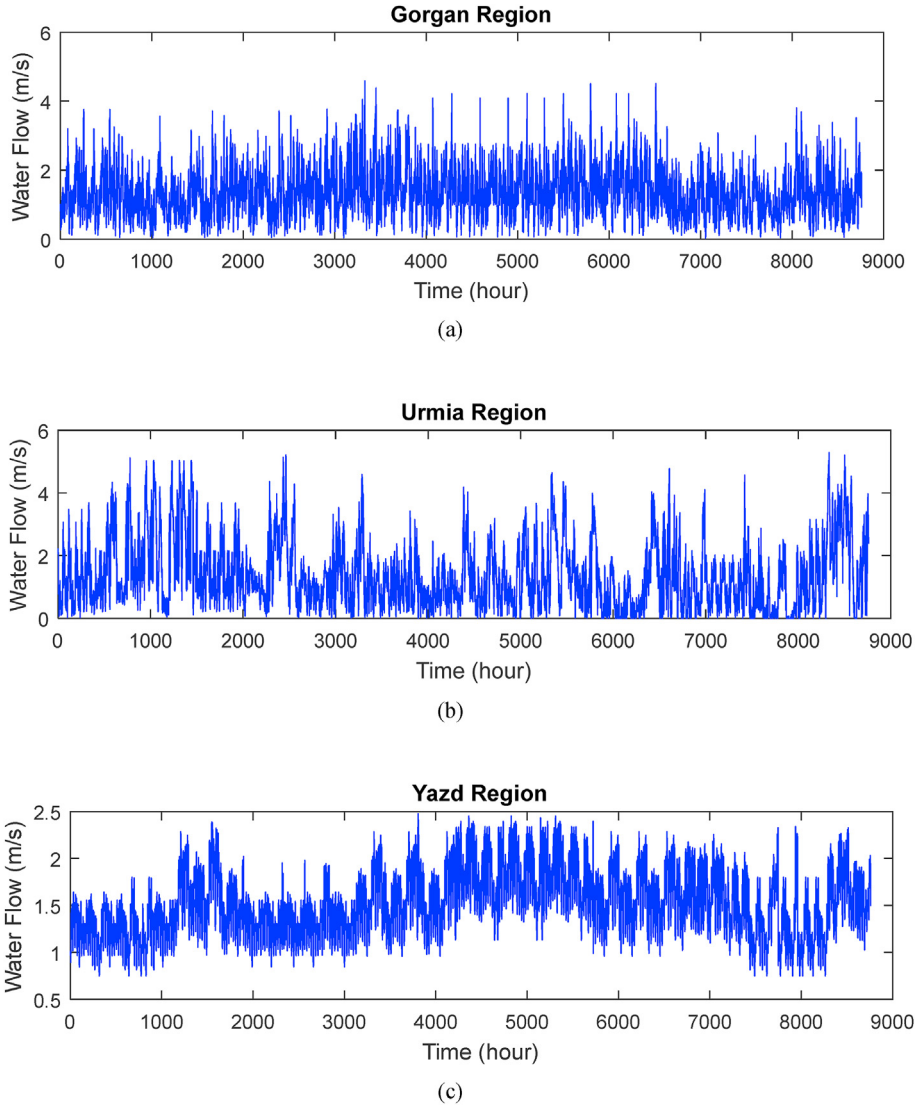


Fig. 8. Annual water flow for different regions a) Gorgan b) Urmia c) Yazd ("Iranian Renewable Energy Organization (SUNA) Tehran, Iran," and "Meteorological Organization, Iran,").

results for the two PV/FC and WT/FC systems shows that the COE values for each of these systems are $1.7924 and $2.8967, respectively, and that the LDP values are 0.0098 and 0.01, respectively. This indicates a better performance from the combination with PV power than with WT power in this region. According to the results of the PV/WT/FC combination, the COE and LDP values are $1.5748 and 0.0096, respectively, indicating that using PV and WT power together with an FC reduces costs and improves load reliability. In addition, the results showed a good performance for the tidal/FC system, with COE and LDP values of $0.8168 and 0.0095, respectively. Comparison with PV/tidal/FC and WT/tidal/FC systems shows that adding PV and WT resources to tidal energy sources reduces costs, and further improves reliability. In addition, adding PV power to tidal power in this region provides lower costs and better reliability. The results also indicated that the best combination for the Gorgan region is a PV/WT/tidal/FC system, with COE and LDP values of $0.7789 and 0.0090, respectively. Therefore, a PV/WT/tidal/FC system is a cost-effective combination for stand-alone applications in the Gorgan region. The optimum power rates for the tidal, PV, WT, EL, FC, and INV sources are 430, 91, 29, 61.47, 36.66,

Table 1
Parameters of hybrid system components (Ahmadi and Abdi, 2016; Moghaddam et al., 2019).

Item	Capital cost ($/unit)	Replacement cost ($/unit)	Maintenance cost ($/unit-yr)	Nominal capacity	Efficiency (%)	Lifetime (Year)
Wind turbine (WT)	3200	700	100	1 kW	–	20
Photovoltaic (PV)	2000	500	33	1 kW	–	20
Tidal	25,000	17,000	100	10 kW	–	20
Electrolyzer (EL)	20,000	1400	100	3 kW	74	5
Hydrogen Tank	1300	200	25	1 kg	95	20
Fuel cell (FC)	20,000	1400	100	3 kW	50	5
Inverter (INV)	800	200	8	1 kW	90	15

Table 2
Sizing results for different systems and regions.

System	Region	$P_{Tidal}(kW)$	$P_{PV}(kW)$	$P_{WT}(kW)$	$P_{EL}(kW)$	$M_{Tank}(kg)$	$P_{FC}(kW)$	$P_{INV}(kW)$
PV/FC	Gorgan	–	1235	–	265.26	142.19	46.26	53.39
	Urmia	–	1883	–	194.76	115.90	44.79	53.42
	Yazd	–	549	–	139.80	30.75	41.94	53.63
WT/FC	Gorgan	–	–	923	369.84	1154.75	38.94	52.10
	Urmia	–	–	1796	813.90	726.12	45.00	52.07
	Yazd	–	–	497	135.84	1395.85	40.29	53.29
Tidal/FC	Gorgan	540	–	–	72.99	135.48	43.83	53.88
	Urmia	600	–	–	132.57	648.34	42.39	53.80
	Yazd	350	–	–	24.93	172.74	29.76	54.61
PV/WT/FC	Gorgan	–	823	234	183.57	103.45	42.06	53.20
	Urmia	–	1510	171	164.64	128.35	45.18	53.32
	Yazd	–	449	28	127.08	40.16	41.52	53.54
PV/Tidal/FC	Gorgan	450	107	–	68.55	93.94	37.80	54.57
	Urmia	330	424	–	79.50	369.67	37.29	53.55
	Yazd	260	92	–	22.95	48.25	28.38	54.64
WT/Tidal/FC	Gorgan	510	–	33	67.26	101.08	41.67	53.79
	Urmia	560	–	42	124.95	606.87	41.55	53.45
	Yazd	340	–	34	28.14	58.55	27.69	54.09
PV/WT/Tidal/FC	Gorgan	430	91	29	61.47	97.96	36.66	55.20
	Urmia	320	425	14	77.70	358.99	36.93	54.98
	Yazd	240	93	14	21.57	44.70	25.53	55.20

and 55.20, respectively, based on the PV/WT/tidal/FC system.

6.2.2. Urmia region

This section discusses the design results for the Urmia region. For this region, the COE values for the PV/FC and WT/FC combinations are $2.0893 and $5.0024, respectively, and the LDP value for both combinations is 0.01. The results showed that load supply-based PV power is more economical than WT power. The PV/WT/FC system showed COE and LDP values of $1.9762 and 0.0096, respectively, indicating that the use of PV power and WT power with an FC further reduces the cost of the load supply. In addition, the results for the PV/tidal/FC and WT/tidal/FC systems showed that the tidal/FC system with PV power (COE and LDP values of $1.0899 and 0.0094, respectively) is more affordable and reliable than a tidal/FC system with WT power (COE and LDP values of $1.2652 and 0.0095, respectively). The results indicate that the best combination for the Urmia region is a hybrid PV/WT/tidal/FC system, with COE and LDP values equal to $1.0864 and 0.0093, respectively. The optimum power rates for the tidal, PV, WT, EL, FC, and INV sources are 320, 425, 14, 77.70, 36.9,3 and 54.98, respectively, based on the hybrid PV/WT/tidal/FC system.

Table 3
Results of reliability and cost for different hybrid system and regions.

System	Region	*LDP*	*HSNPC (M$)*	*COE ($/kWh)*
PV/FC	Gorgan	0.0098	5.655	1.7924
	Urmia	0.01	6.592	2.0893
	Yazd	0.0097	2.826	0.8968
WT/FC	Gorgan	0.01	9.103	2.8967
	Urmia	0.01	15.752	5.0024
	Yazd	0.0099	5.810	1.8466
Tidal/FC	Gorgan	0.0095	2.573	0.8168
	Urmia	0.0097	4.022	1.2797
	Yazd	0.0094	1.670	0.5302
PV/WT/FC	Gorgan	0.0096	4.964	1.5748
	Urmia	0.0096	6.240	1.9762
	Yazd	0.0095	2.630	0.8343
PV/Tidal/FC	Gorgan	0.0092	2.474	0.7854
	Urmia	0.0094	3.433	1.0899
	Yazd	0.0091	1.433	0.4546
WT/Tidal/FC	Gorgan	0.0094	2.547	0.8084
	Urmia	0.0095	3.977	1.2652
	Yazd	0.0092	1.615	0.5129
PV/WT/Tidal/FC	Gorgan	0.0090	2.454	0.7789
	Urmia	0.0093	3.421	1.0864
	Yazd	0.0090	1.412	0.4477

6.2.3. Yazd region

This section presents the design results for the Yazd region. The COE values for the PV/FC and WT/FC combinations are $0.8968 and $1.8466, respectively. The LDP values are 0.0097 and 0.0099, respectively. The results indicated that supplying the system load using PV power is more economic than using WT power. The PV/WT/FC system results showed COE and LDP values equal to $0.8343 and 0.0095, respectively, resulting in a lower cost and better reliability when using the contribution of PV and WT resources with an FC. The results of the PV/tidal/FC and WT/tidal/FC systems confirmed that a tidal/FC system with PV power (COE and LDP values of $0.4546 and 0.0091, respectively) is more economical and reliable than a tidal/FC system with WT power (COE and LDP values of $0.5129 and 0.0092, respectively). The best and optimal combination for the Yazd region is a PV/WT/tidal/FC system, with COE and LDP values equal to $0.4477 and 0.0090, i.e. superior to the other combinations. The optimum power rates for the tidal, PV, WT, EL, FC, and INV sources are 240, 93, 14, 21.57, 25.53, and 55.20, respectively, based on the PV/WT/tidal/FC system.

6.2.4. Results comparison for different regions

Comparisons of the HSNPC and COE values for different combinations of hybrid systems and under different study regions are shown in Figs. 9 and 10, respectively. As can be seen from the analysis of the results in the previous sections, the PV/WT/tidal/FC combination (optimal) has lower HSNPC and COE values. Therefore, the best economic combination for supplying the load in these regions is the PV/WT/tidal/FC system, based on the patterns of irradiance, wind speed, and water flow. The results showed that the COE values for the Gorgan, Urmia, and Yazd regions were $0.7789, $1.0864 and $0.4477, respectively. Thus, for supplying a similar load demand, Yazd has a lower COE, and Urmia has a higher COE.

6.2.5. Unit contributions for different regions

In this section, the power output curves of each unit, including the PV, WT, tidal and FC units in the optimal hybrid PV/WT/tidal/FC combinations in different regions, are presented in Figs. 11–17. In addition, Figs. 11–17 illustrate the energy management of the units (including the generating units, as well as the FC power production to compensate for the hybrid power deficit) and the load reliability improvement for the first 168 h of simulation (1 week). The figures

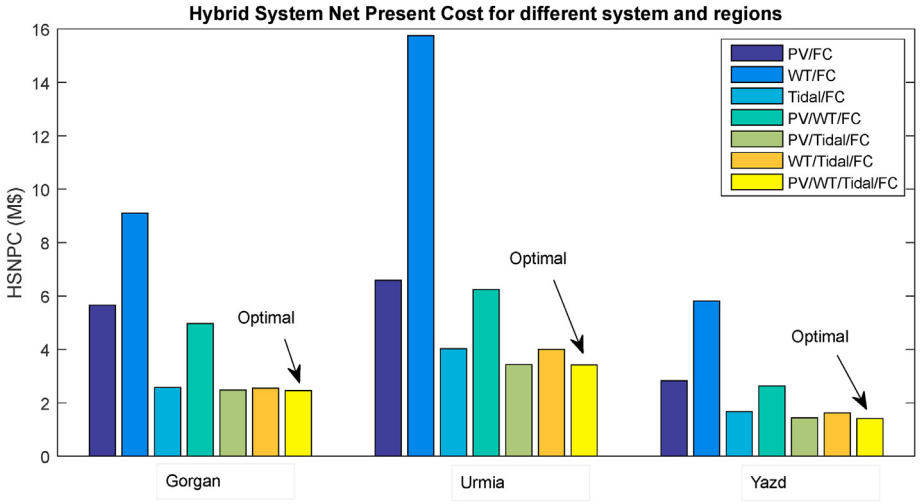


Fig. 9. Hybrid system net present cost (HSNPC) for different systems and regions.

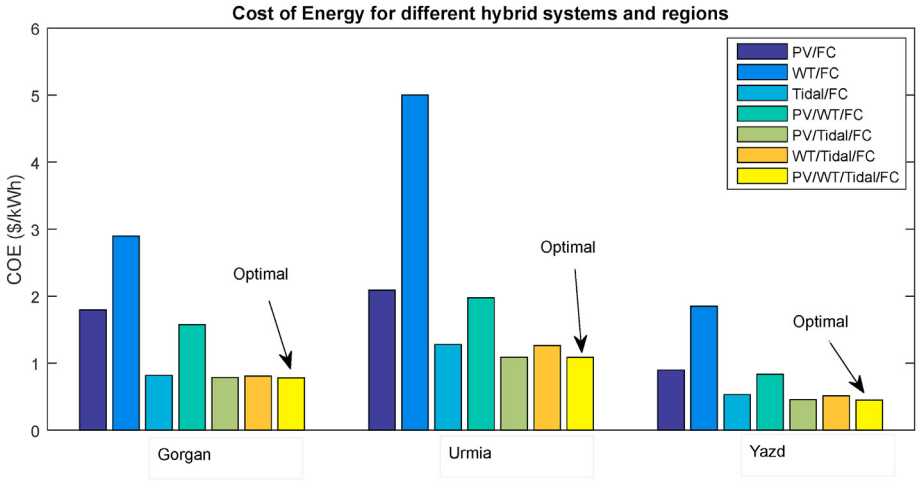


Fig. 10. Cost of energy (COE) for different systems and regions.

illustrate the support of the load and the improvement of its reliability owing to the storage system, especially when using the FC to compensate for a power deficit in the hybrid system. PV and WT power variation for PV/WT/tidal/FC system and different regions are illustrated in Figs. 11 and 12. These figures showed that the PV contribution is more for Urmia and WT contribution is more for Gorgan. Figs. 13 and 14 showed that the Tidal power is more for the Gorgan region and FC power is more for Urmia region compared to

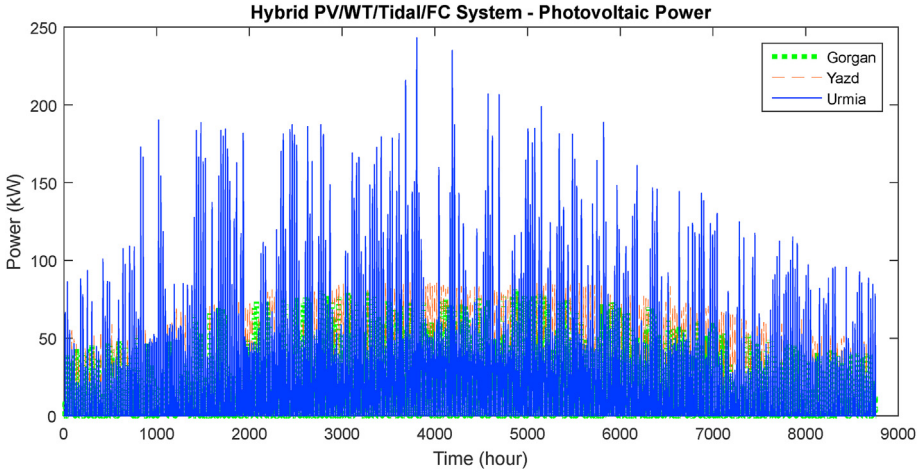


Fig. 11. PV power variation for PV/WT/tidal/FC system and different regions.

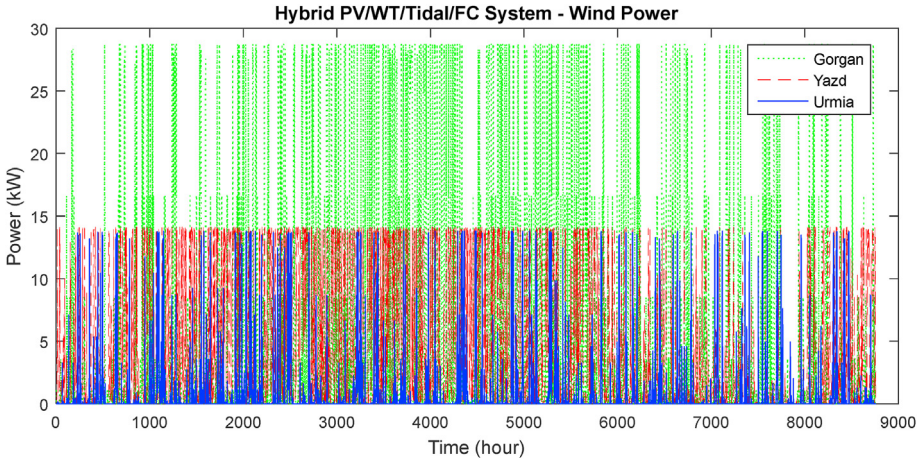


Fig. 12. WT power variation for PV/WT/tidal/FC system and different regions.

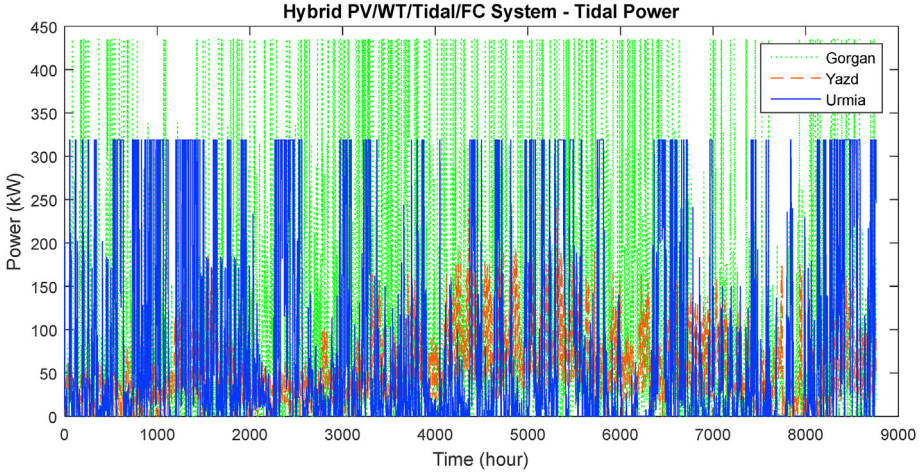


Fig. 13. Tidal power variation for PV/WT/tidal/FC system and different regions.

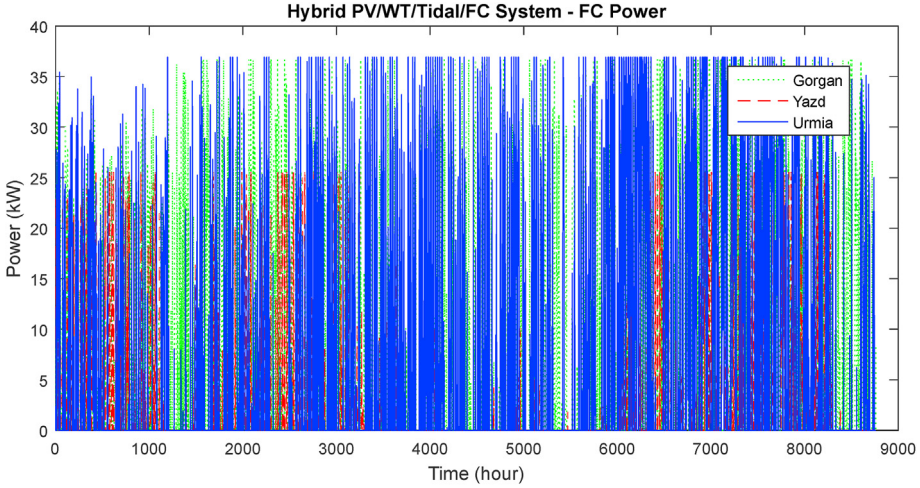


Fig. 14. FC power variation for PV/WT/tidal/FC system and different regions.

different regions, respectively. Also, energy management and unit contributions for PV/WT/tidal/FC system for Urmia region for different regions are shown in Figs. 15–17. In these figures illustrated that for different regions the storage system (FC power) is supported the continuous supply the load demand to achieve the optimal reliability.

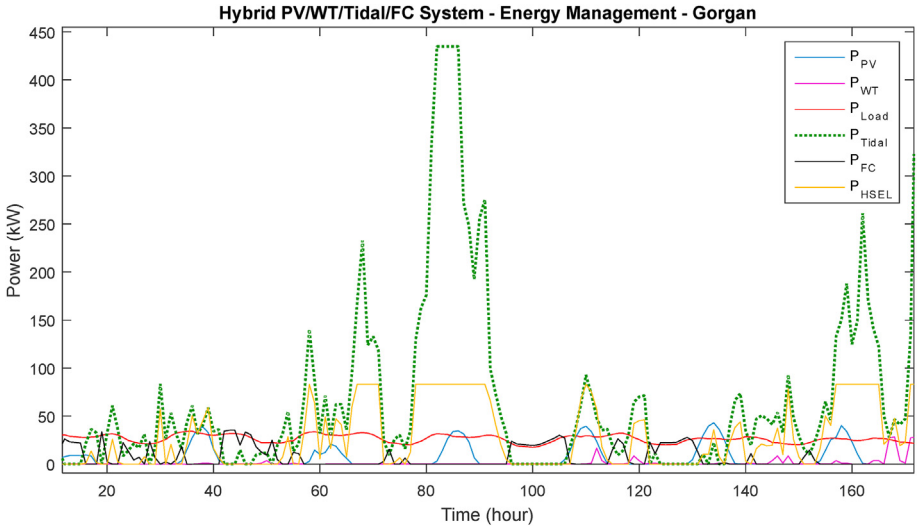


Fig. 15. Energy management and unit contributions for PV/WT/tidal/FC system for Gorgan region.

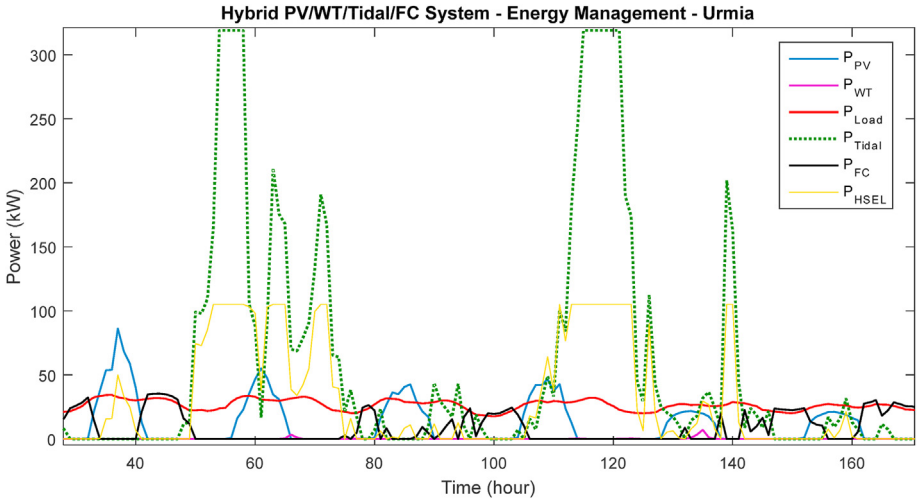


Fig. 16. Energy management and unit contributions for PV/WT/tidal/FC system for Urmia region.

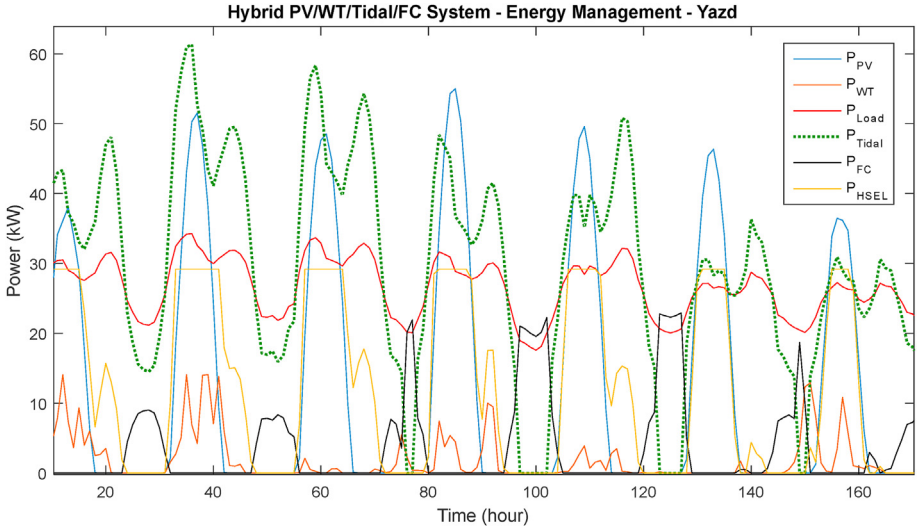


Fig. 17. Energy management and unit contributions for PV/WT/tidal/FC system for Yazd region.

6.2.6. HS energy for different regions

The HS energy curve for the PV/WT/tidal/FC system is plotted in Fig. 18 for the various regions for one year. It is observed that the Yazd region has the least dependence on the storage system, owing to its good potential for renewable energies, especially PV energy, which reduces system costs. In contrast, the most dependence on the storage system is in the Urmia region, owing to the inadequate potential for wind and radiation. Therefore, according to Tables 2

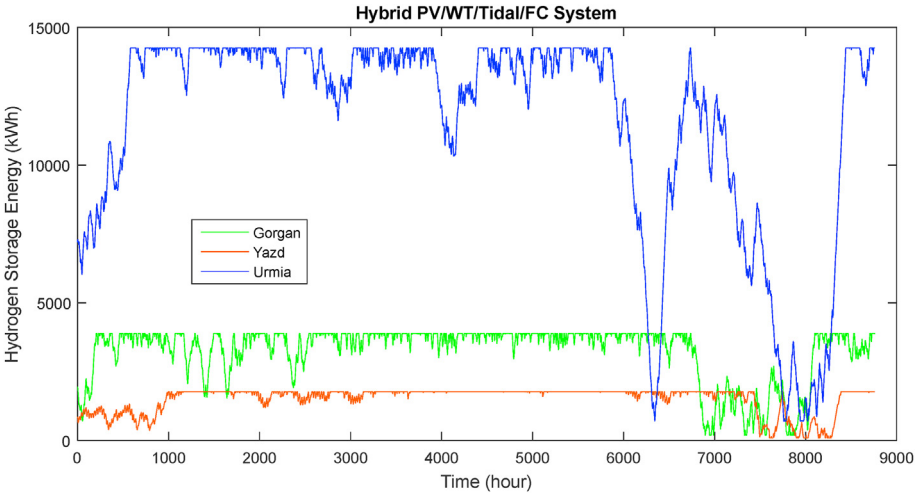


Fig. 18. Hydrogen tank energy variation for PV/WT/tidal/FC system and different regions.

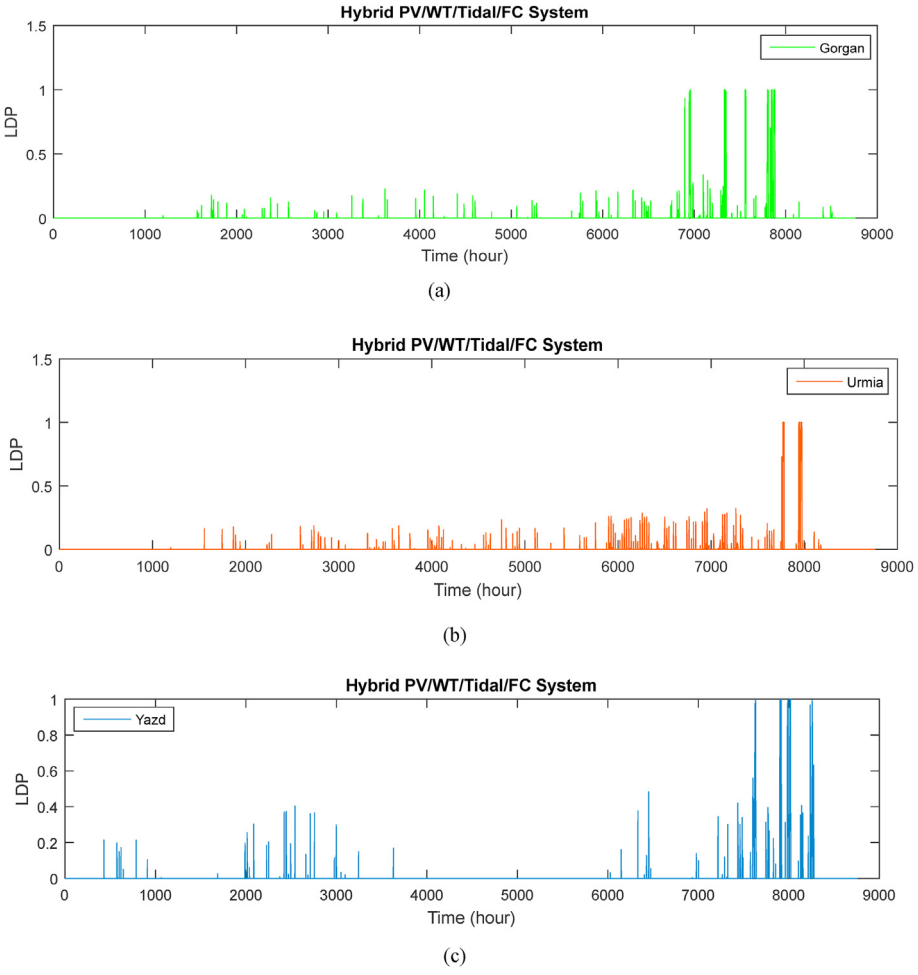


Fig. 19. Load deficit probability (LDP) variation for PV/WT/tidal/FC system and different regions.

and 3 and the results presented in this section, it is observed that increasing the level of dependence on the storage system increases the COE for supplying the load demand.

6.2.7. Load deficit probability (LDP) variations for different regions

In this study, the design of the hybrid systems is performed for $LDP^{max} = 5\%$. In other words, a hybrid system will not be able to meet the total load demand, owing to inherent fluctuations in the irradiance, wind speed, and water flow (for PV/WT/tidal/FC system). In Fig. 19, the LDP variation curve for the different regions is plotted. According to Table 3, the values of the LDP for Gorgan, Urmia, and Yazd are 0.0090, 0.0093, and 0.0090, respectively. Moreover, a higher load reliability is confirmed for the Gorgan and Yazd regions.

6.2.8. Superiority of WOA compared with particle swarm optimisation (PSO)

In this section, the superiority of the WOA over PSO for the optimal combination in hybrid system design is proven, based on the reduced HSNPC and LDP values. The convergence curves of WOA and PSO in the optimal combination designs for different regions are illustrated in Figs. 20–22. In addition, the comparative results for all regions are presented in Tables 4 and 5.

As can be seen, the WOA method converges faster than the PSO method, or in less iteration, it is able to converge and find the optimal solution. Also, the convergence tolerance of the WOA method is less than the PSO method. So the WOA is more precise. In terms of numerical results, according to Table 4, the WOA method has achieved HSNPC, lower COE and better (less) LDP, which indicates the superiority of the proposed method.

6.2.9. Comparison with previous studies

A comparison with previous studies is presented in Table 6. In this table, the COE per kW is presented for different previous studies with different methods and regions. However, the amount of cost varies for each geographic region, depending on the patterns of radiation and wind speed. Therefore, less or more cost does not in any way imply the superiority or weakness of the proposed

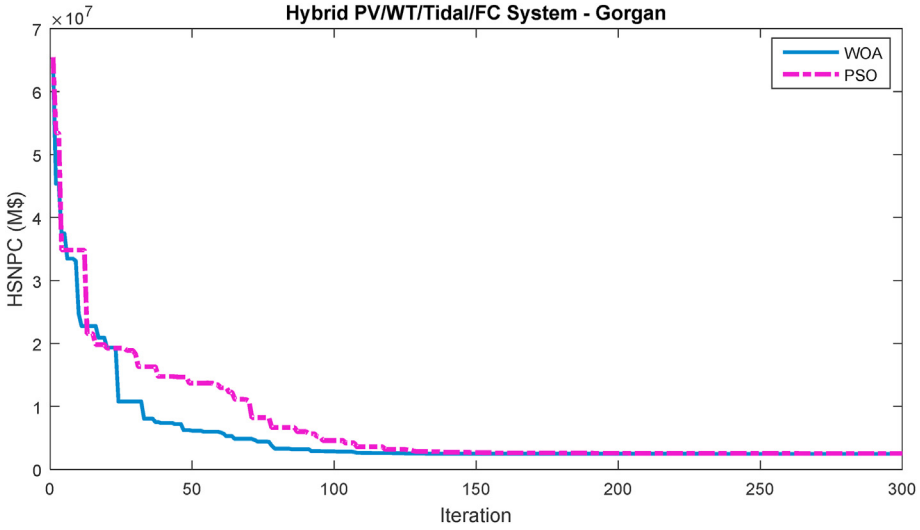


Fig. 20. Convergence curve of WOA and particle swarm optimisation (PSO) for PV/WT/tidal/FC system design for Gorgan region.

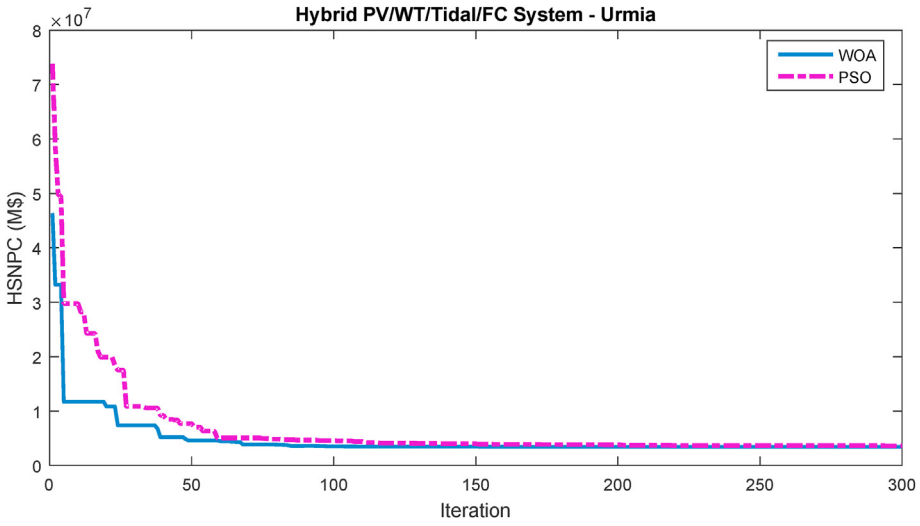


Fig. 21. Convergence curve of WOA and PSO for PV/WT/tidal/FC system design for Urmia region.

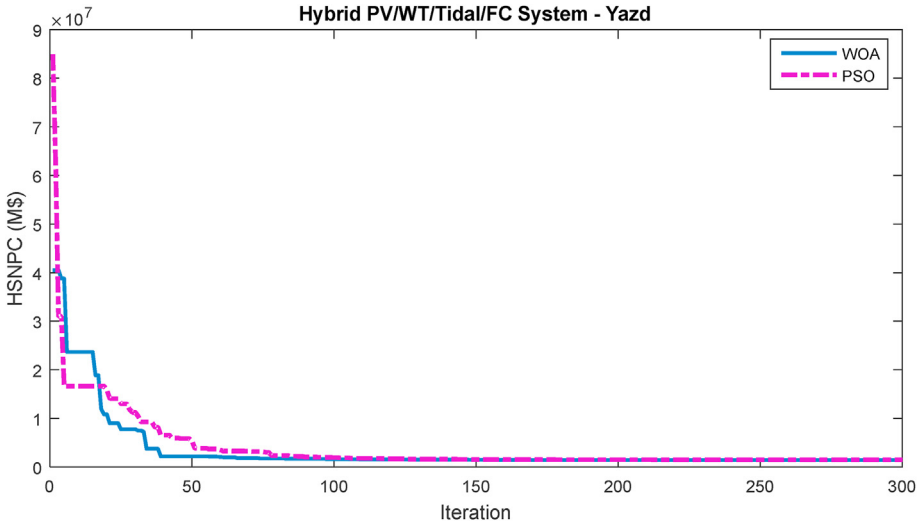


Fig. 22. Convergence curve of WOA and PSO for PV/WT/tidal/FC system design for Yazd region.

Table 4
Results of optimal sizing for photovoltaic (PV)/wind turbine (WT)/tidal/fuel cell (FC) in different regions for whale optimisation algorithm (WOA) particle swarm optimisation (PSO).

Region	Method	$P_{Tidal}(kW)$	$P_{PV}(kW)$	$P_{WT}(kW)$	$P_{EL}(kW)$	$M_{Tank}(kg)$	$P_{FC}(kW)$	$P_{INV}(kW)$
Gorgan	WOA	430	91	29	61.47	97.96	36.66	55.20
	PSO	490	8	36	67.14	102.24	40.38	47.04
Urmia	WOA	320	425	14	77.70	358.99	36.93	54.98
	PSO	240	462	66	66.03	437.83	41.40	55.20
Yazd	WOA	240	93	14	21.57	44.70	25.53	55.20
	PSO	220	94	25	12.45	166.53	20.07	52.89

Table 5
Results for reliability (load deficit probability, LDP) and cost (cost of energy (COE) and hybrid system net present cost (HSNPC)) for PV/WT/tidal/FC and different regions.

System	Region	*LDP*	*HSNPC (M$)*	*COE ($/kWh)*
Gorgan	WOA	0.0090	2.454	0.7789
	PSO	0.0094	2.526	0.8018
Urmia	WOA	0.0093	3.421	1.0864
	PSO	0.0097	3.621	1.1482
Yazd	WOA	0.0090	1.412	0.4477
	PSO	0.0092	1.476	0.4498

method, but rather the potential of that region in terms of the cost of the proposed hybrid systems. This cost will inform energy system designers in the economic evaluation of a renewable power plant, based on the radiation and wind speed potential of the region for implementing the system. The COE values for Gorgan, Urmia, and Yazd are $0.7789, $1.0864, and $0.4477.

7. Conclusion

In this study, a hybrid PV/WT/tidal/FC clean energy system design is evaluated for three regions of Iran, i.e. Gorgan, Urmia, and Yazd. Real data of the radiation and wind speed in these regions is applied to develop cost-effective sizing of the hybrid system components. The objective function is considered as minimisation of the HSNPC and satisfying a reliability constraint (LDP). The WOA is applied to find the best combination of reliable and cost-effective systems in view of the cost/reliability evaluation. Different

Table 6
Comparison with previous studies.

Ref	Hybrid system type	Region	Cost of energy ($/kWh)
Fazelpour et al. (2016)	PV/Wind/FC	Tehran/Iran	0.83
Ashourian et al. (2013)	PV/Wind/Battery	Tioman/Malaysia	1.10
Moghaddam et al. (2019)	PV/WT/FC	Ardabil/Iran	0.52
Borhanazad et al. (2014)	PV/WT/Diesel/Battery	Nahavand/Iran	1.87
Borhanazad et al. (2014)	PV/WT/Diesel/Battery	Rafsanjan/Iran	0.32
Borhanazad et al. (2014)	PV/WT/Diesel/Battery	Khash/Iran	0.35
Al-Sharafi et al. (2017)	PV/FC/Battery	Jeddah, Saudi Arabia	1.40
Dursun (2012)	PV/Diesel/Battery	Kırklareli/Turkey	0.81
This paper	PV/WT/Tidal/FC	Gorgan	0.7789
This paper	PV/WT/Tidal/FC	Urmia	1.0864
This paper	PV/WT/Tidal/FC	Yazd	0.4477

combinations of a hybrid system consisting of tidal, PV, WT, and FC-based hydrogen power are optimally designed, and the best combination with the lowest cost of energy (COE) and load deficit probability (LDP) is found for each region. The results showed that the application of all sources as PV/WT/tidal/FC is optimal for supplying the load in all regions. The results showed that the contribution of all resources, as combined with an HS system, is capable of improving the load reliability. The obtained results showed that the COE values for Gorgan, Urmia, and Yazd are $0.7789, $1.0864, and $0.4477 and that the LDP values for these regions are 0.9, 0.93, and 0.0.9 for PV/WT/tidal/FC system. Tidal power had a greater contribution to the Gorgan and Yazd regions, and PV power had a greater contribution to the Urmia region. Also, less contribution is found by wind power in all regions. The superiority of the WOA over PSO is proved by obtaining less cost and better reliability for all regions.

CRediT authorship contribution statement

Amirreza Naderipour: Conceptualization, Methodology. **Zulkurnain Abdul-Malek:** Supervision, Project administration. **Saber Arabi Nowdeh:** Writing - original draft, Software. **Hesam Kamyab:** Data curation. **Amir Reza Ramtin:** Resources. **Saman Shahrokhi:** Visualization. **Jiří Jaromír Klemeš:** Supervision, Writing - review & editing, Proofreading.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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